Project: Worked examples
Title: Beam formulas (single span)
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Closed-form reactions, maximum moment and its position, maximum shear and maximum deflection for the standard single-span beam cases, as a quick check. Left end at x = 0; loads downward; sagging moments positive; deflection downward positive.
Roark Table 8.1 formulas in own notation. For the simply supported point load at a (b = L − a): R_A = Pb/L, M_max = Pab/L, δ_max = Pb(L² − b²)^(3/2)/(9√3 L EI) at x = √((L² − b²)/3) when a ≥ b. For the propped cantilever: R_B = 3wL/8, M_A = −wL²/8, sag M_max = 9wL²/128 at x = 5L/8.
Ref: Roark's Formulas for Stress and Strain Ch. 8, Table 8.1 — Shear, moment, slope and deflection formulas for elastic straight beams

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
case	Case	simply supported, point load at a		assumption: Single span, pin–roller.	any
L	Span	6	m	drawing: DRG-FL-201 rev A	0.1 … 100 m
E	Modulus of elasticity	210	GPa	spec: BS EN 1993-1-1 3.2.6	1 … 500 GPa
I	Second moment of area	8503	cm^4	reference: UB 305×165×40 (Blue Book)	1 … 100000000 cm^4
P	Point load (downward)	50	kN	spec: Load schedule LS-01: plant load	0 … 100000 kN
a	Load position from the left end	4	m	drawing: DRG-FL-201 rev A	0 … 100 m

3. Assumptions

A1. Prismatic, linear elastic, Euler–Bernoulli beam; small deflections. (Roark Table 8.1 conditions.)
A2. A cantilever is fixed at the left end (x = 0) and free at the right. (Case convention.)
A3. case (Case) is taken as simply supported, point load at a. (Single span, pin–roller.)

4. Working

[5] EI = E * I
= 210 GPa × 8503 cm⁴
= 17856 kN·m²
Flexural rigidity

[8] b = L - a
= 6 m − 4 m
= 2 m
Distance from the load to the right end

```
[9] R_A = P * b / L
= 50 kN × 2 m / 6 m
= 16.667 kN
```
Roark 8.1 2a

Left reaction

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[10] R_B = P * a / L
= 50 kN × 4 m / 6 m
= 33.333 kN
```
Right reaction

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[11] M_max = P * a * b / L
= 50 kN × 4 m × 2 m / 6 m
= 66.667 kN·m
```
Maximum moment, under the load

```
[12] x_M = a
= 4 m
= 4 m
```
Position of the maximum moment

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[13] M_min = 0 kN*m
= 0 kN·m
```
Minimum moment

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[14] V_max = max(R_A, R_B)
= max(16.667 kN, 33.333 kN)
= 33.333 kN
```
Maximum shear

```
[15] c_far = b
= 2 m
= 2 m
```
Shorter segment (the deflection formula's b)

```
[16] x_delta = sqrt((L^2 - c_far^2) / 3)
= sqrt(((6 m)^2 - (2 m)^2) / 3)
= 3.266 m
```
Position of the maximum deflection

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[17] delta_max = P * c_far * sqrt((L^2 - c_far^2)^3) / (9 * sqrt(3) * L * EI)
= 50 kN × 2 m × sqrt(((6 m)^2 - (2 m)^2)^3) / (9 × sqrt(3) × 6 m × 17856 kN·m²)
= 10.839 mm
```
Maximum deflection

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	16.667 kN	Left reaction
R_B	33.333 kN	Right reaction
M_max	66.667 kN·m	Maximum sagging moment
x_M	4 m	Position of M_max
M_min	0 kN·m	Minimum (hogging) moment
V_max	33.333 kN	Maximum shear
delta_max	10.839 mm	Maximum deflection
x_delta	3.266 m	Position of δ_max

6. Checks (governing first)

No pass/fail checks in this calculation.

7. Applicability

No warnings.

8. Verification

```
Template struct.beam-formulas@1.0.0 Content hash 031ca2bf31f2f497 Engine calc-engine/1.0.0
Known-answer cases passed 8/8 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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